Pranav Ray

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Profile

Full-stack developer and systems programmer building production web applications and distributed systems tooling in Go, from P2P networking and gossip protocols to real-time web platforms with 150+ active users.

Projects

hotaru: Raft Consensus Implementation GitHub *(in progress)*

- Implementing the Raft consensus algorithm from scratch in Go, covering leader election via randomized timeouts, log replication across cluster nodes, and commit guarantees backed by majority acknowledgment.
- Building a replicated key-value store on top of the Raft layer as the state machine, exposing GET/SET/DELETE via REST API with automatic leader redirection for writes.
- Engineering a Next.js + D3 dashboard visualizing live leader elections, log entry propagation, commit index advancement, and node failure/recovery in real time via WebSocket stream.

Tech: Go, Raft, net/rpc, WebSockets, Next.js, TypeScript, Tailwind CSS, D3.js

tobira: Distributed Rate Limiter GitHub

- Built a distributed rate limiter in Go coordinated via a custom gossip protocol over UDP, no Redis, no central coordinator, with nodes discovering and syncing state directly with each other.
- Implemented state-based CRDTs (grow-only counter maps) for deterministic conflict-free state merging, ensuring global rate limit counts converge correctly even after network partitions and node failures.
- Engineered four pluggable rate limiting algorithms (fixed window, sliding window, token bucket, leaky bucket) hot-swappable at runtime, with heartbeat-based failure detection and a Next.js + D3 topology dashboard.

Tech: Go, UDP, Gossip Protocol, CRDT, WebSockets, Next.js, TypeScript, D3.js, Docker

tsuna: P2P Video Sync CLI GitHub

- Implemented raw UDP hole punching and STUN-based NAT traversal in Go from scratch, enabling two peers behind separate home routers to connect directly with no media server involvement.
- Engineered a custom NTP-style clock reconciliation algorithm targeting sub-100ms playback sync across peers, using continuous round-trip time compensation to calculate and correct clock offset between machines.
- Built a lightweight WebSocket signaling server for initial peer handshake coordination, the only server in the system, going hands-off permanently after the hole-punch exchange completes.

Tech: Go, UDP/P2P, STUN, WebSockets, Cobra, BubbleTea, Next.js, TypeScript, Docker, Railway

typingterminal: Typing Practice Platform Live

- Engineered and shipped a retro terminal-style typing platform to **150+ active users** and **3,000+ site visits** with zero third-party auth or backend infrastructure.
- Architected real-time multiplayer race rooms with live leaderboards using Convex's reactive query engine, handling concurrent session state across multiple clients without explicit WebSocket management.
- Built per-user performance analytics tracking WPM, accuracy curves, and historical progress with session-level granularity.

Tech: React, Next.js, TypeScript, Tailwind CSS, Convex, Framer Motion

Technical Skills

Languages: TypeScript, JavaScript, Go, Python, C/C++, Swift
Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Framer Motion, GSAP, Three.js, SwiftUI
Backend: Node.js, Bun, Express.js, Convex, Prisma, REST APIs, WebSockets
Databases: MongoDB, PostgreSQL, Redis, Firebase, Supabase
Distributed Systems: Gossip Protocols, CRDTs, Raft Consensus, UDP Networking, NAT Traversal
Tools: Git, Vercel, Docker, Railway, Figma, Postman, Leaflet, Cobra, BubbleTea, Xcode

Education

Manipal University Jaipur, Jaipur, India
B.Tech in Computer Science 2024 – 2028
CGPA: 8.0 / 10